

## SAFETY DATA SHEET

Revision Date 01-Apr-2024

Revision Number 3

### 1. Identification

**Product Name** Calcium zirconium isopropoxide, 10% w/v in isopropanol

**Cat No. :** 41971

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |            |
|--------------------------------------------------|------------|
| Flammable liquids                                | Category 2 |
| Skin Corrosion/Irritation                        | Category 2 |
| Serious Eye Damage/Eye Irritation                | Category 1 |
| Specific target organ toxicity (single exposure) | Category 3 |
| Target Organs - Central nervous system (CNS).    |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor  
Causes skin irritation  
Causes serious eye damage  
May cause drowsiness or dizziness



### Precautionary Statements

#### Prevention

Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Keep cool

#### Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

#### Skin

If skin irritation occurs: Get medical advice/attention  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

#### Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER or doctor/physician

#### Fire

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

#### Storage

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

None identified

## 3. Composition/Information on Ingredients

| Component                      | CAS No      | Weight % |
|--------------------------------|-------------|----------|
| Isopropyl alcohol              | 67-63-0     | 90.00    |
| Calcium zirconium isopropoxide | 334512-82-2 | 10.00    |

## 4. First-aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                            |                                                                                                                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                        |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                             |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes eye burns. Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                    |

## 5. Fire-fighting measures

|                                         |                                                   |
|-----------------------------------------|---------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                          |
| <b>Flash Point</b>                      | 12 °C / 53.6 °F                                   |
| <b>Method -</b>                         | No information available                          |
| <b>Autoignition Temperature</b>         | No information available                          |
| <b>Explosion Limits</b>                 |                                                   |
| <b>Upper</b>                            | No data available                                 |
| <b>Lower</b>                            | No data available                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                          |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

None known.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 3                   | 0                  | -                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                            |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage.**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component         | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH                                                                                                         | Mexico OEL (TWA)              |
|-------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------|
| Isopropyl alcohol | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
OSHA - Occupational Safety and Health Administration  
NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

### Personal Protective Equipment

#### Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

low boiling organic solvent. Type AX. Brown. conforming to EN371. or. Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                          |
|----------------------------------|--------------------------|
| Physical State                   | Liquid                   |
| Appearance                       | No information available |
| Odor                             | Alcohol                  |
| Odor Threshold                   | No information available |
| pH                               | No information available |
| Melting Point/Range              | No data available        |
| Boiling Point/Range              | No information available |
| Flash Point                      | 12 °C / 53.6 °F          |
| Evaporation Rate                 | No information available |
| Flammability (solid,gas)         | Not applicable           |
| Flammability or explosive limits |                          |
| Upper                            | No data available        |

|                                        |                          |
|----------------------------------------|--------------------------|
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C18 H42 CaO6 Zr          |
| Molecular Weight                       | 485.83                   |

## 10. Stability and reactivity

|                                  |                                                                   |
|----------------------------------|-------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                        |
| Stability                        | Stable under normal conditions.                                   |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents                                           |
| Hazardous Decomposition Products | None under normal use conditions                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                          |
| Hazardous Reactions              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Vapor LC50  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

#### Component Information

| Component         | LD50 Oral                                  | LD50 Dermal         | LC50 Inhalation       |
|-------------------|--------------------------------------------|---------------------|-----------------------|
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat ) | 72.6 mg/L ( Rat ) 4 h |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                      | CAS No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------|-------------|------------|------------|------------|------------|------------|
| Isopropyl alcohol              | 67-63-0     | Not listed | Not listed | Not listed | Not listed | Not listed |
| Calcium zirconium isopropoxide | 334512-82-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                       |                           |
|-----------------------|---------------------------|
| Mutagenic Effects     | No information available  |
| Reproductive Effects  | No information available. |
| Developmental Effects | No information available. |

|                                            |                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Teratogenicity                             | No information available.                                                                                           |
| STOT - single exposure                     | Central nervous system (CNS)                                                                                        |
| STOT - repeated exposure                   | None known                                                                                                          |
| Aspiration hazard                          | No information available                                                                                            |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| Endocrine Disruptor Information            | No information available                                                                                            |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                      |

## 12. Ecological information

### Ecotoxicity

| Component         | Freshwater Algae                                                                                           | Freshwater Fish                                                                                                                                                                                                          | Microtox                                              | Water Flea                                      |
|-------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------|
| Isopropyl alcohol | EC50: > 1000 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h<br>(Desmodesmus subspicatus) | LC50: = 9640 mg/L, 96h<br>flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h<br>(Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h<br>static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h<br>(Daphnia) | = 35390 mg/L EC50<br>Photobacterium phosphoreum 5 min | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h |

|                               |                           |
|-------------------------------|---------------------------|
| Persistence and Degradability | Persistence is unlikely   |
| Bioaccumulation/ Accumulation | No information available. |
| Mobility                      | No information available. |

| Component         | log Pow |
|-------------------|---------|
| Isopropyl alcohol | 0.05    |

## 13. Disposal considerations

|                        |                                                                                                                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste Disposal Methods | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                      |             |
|----------------------|-------------|
| UN-No                | UN1219      |
| Proper Shipping Name | ISOPROPANOL |
| Hazard Class         | 3           |
| Packing Group        | II          |

### TDG

|                      |             |
|----------------------|-------------|
| UN-No                | UN1219      |
| Proper Shipping Name | ISOPROPANOL |
| Hazard Class         | 3           |
| Packing Group        | II          |

### IATA

|                      |             |
|----------------------|-------------|
| UN-No                | UN1219      |
| Proper Shipping Name | ISOPROPANOL |
| Hazard Class         | 3           |

|                      |             |
|----------------------|-------------|
| Packing Group        | II          |
| IMDG/IMO             |             |
| UN-No                | UN1219      |
| Proper Shipping Name | ISOPROPANOL |
| Hazard Class         | 3           |
| Packing Group        | II          |

## 15. Regulatory information

### United States of America Inventory

| Component                      | CAS No      | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|--------------------------------|-------------|------|--------------------------------------------------|--------------------------------|
| Isopropyl alcohol              | 67-63-0     | X    | ACTIVE                                           | -                              |
| Calcium zirconium isopropoxide | 334512-82-2 | -    | -                                                | -                              |

#### Legend:

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

#### TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)

Not applicable

#### TSCA 12(b) - Notices of Export

Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                      | CAS No      | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|--------------------------------|-------------|-----|------|-----------|-------|------|------|------|-------|----------|
| Isopropyl alcohol              | 67-63-0     | X   | -    | 200-661-7 | X     | X    | X    | X    | X     | KE-29363 |
| Calcium zirconium isopropoxide | 334512-82-2 | -   | -    | -         | -     | -    | -    | -    | -     | -        |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component         | CAS No  | Weight % | SARA 313 - Threshold<br>Values % | SARA 313 - Reporting<br>thresholds |
|-------------------|---------|----------|----------------------------------|------------------------------------|
| Isopropyl alcohol | 67-63-0 | 90.00    | 1.0 %                            | -                                  |

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

OSHA - Occupational Safety and  
Health Administration Not applicable

#### CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive

Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know  
Regulations**

| Component         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------|---------------|------------|--------------|----------|--------------|
| Isopropyl alcohol | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland  
Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH**

| Component                      | CAS No      | REACH (1907/2006) -<br>Annex XIV - Substances<br>Subject to Authorization | REACH (1907/2006) -<br>Annex XVII - Restrictions<br>on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 -<br>Candidate List of<br>Substances of Very High<br>Concern (SVHC) |
|--------------------------------|-------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol              | 67-63-0     | -                                                                         | Use restricted. See item<br>75.<br>(see link for restriction<br>details)               | -                                                                                                                 |
| Calcium zirconium isopropoxide | 334512-82-2 | -                                                                         | -                                                                                      | -                                                                                                                 |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                         | CAS No      | OECD HPV       | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|-----------------------------------|-------------|----------------|---------------------------------|------------------------------|--------------------------------------------------|
| Isopropyl alcohol                 | 67-63-0     | Listed         | Not applicable                  | Not applicable               | Not applicable                                   |
| Calcium zirconium<br>isopropoxide | 334512-82-2 | Not applicable | Not applicable                  | Not applicable               | Not applicable                                   |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component | CAS No | Seveso III Directive<br>(2012/18/EC) - | Seveso III Directive<br>(2012/18/EC) - | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|-----------|--------|----------------------------------------|----------------------------------------|-------------------------------|---------------------------------------|
|-----------|--------|----------------------------------------|----------------------------------------|-------------------------------|---------------------------------------|



|                                   |             | Qualifying Quantities<br>for Major Accident<br>Notification | Qualifying Quantities<br>for Safety Report<br>Requirements |                |                |
|-----------------------------------|-------------|-------------------------------------------------------------|------------------------------------------------------------|----------------|----------------|
| Isopropyl alcohol                 | 67-63-0     | Not applicable                                              | Not applicable                                             | Not applicable | Annex I - Y42  |
| Calcium zirconium<br>isopropoxide | 334512-82-2 | Not applicable                                              | Not applicable                                             | Not applicable | Not applicable |

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 01-Apr-2024  
**Print Date** 01-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**